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# Optical Profilometry Versus the Stylus: A Spectral Theory for Closing the Accuracy Gap

Dawid Kucharski[1,*], Michał Jakubowicz[1]

[1] *Institute of Mechanical Technology, Poznan University of Technology, Jacka Rychlewskiego 1 Street, 61-131 Poznań, Poland*

**Corresponding Author: Dawid Kucharski , e-mail: dawid.kucharski@put.poznan.pl*

***Abstract*— Stylus profilometry and optical interferometry are compared in the spatial-frequency domain through a measurement-operator formalism computed from first-principles instrument physics. The stylus channel is modelled as morphological tip dilation with raster sampling; its empirical transfer is approximately flat with a curvature-bias excess, crossing 3 dB only near the sampling scale. The optical channel follows the incoherent imaging transfer function with a diffraction cutoff of $2000\,\mathrm{mm}^{-1}$. A $k$-space equivalence regime covering $9.6\,\% \pm 0.3\,\%$ of the measured bandwidth is identified at a crossover of $454/\mathrm{mm} \pm 166/\mathrm{mm}$. The optical model reproduces measured white-light-interferometry transfer functions from a public dataset, and all results are checked by eighteen automated verification tests. The study is computational and provides a falsifiable experimental roadmap.**

***Keywords*—Surface metrology; instrument transfer function; stylus profilometry; white-light interferometry; power spectral density**

## I. INTRODUCTION

Surface topography measurement underpins quality control in precision manufacturing, functional surface design in tribology, and performance verification in optical fabrication. Two modalities dominate practice: contact stylus profilometry and non-contact optical interferometry, in particular white-light interferometry (WLI). Comparative studies date back three decades [1], and power spectral density (PSD) analysis has since become the standard for quantitative, scale-resolved surface characterisation [2], in line with the paradigm shift towards information-rich surface metrology [3].

The existing comparative literature relies almost exclusively on scalar metrics such as the root-mean-square (rms) height error or areal roughness *Sa*. Such metrics collapse all spatial-frequency information into a single number: a stylus with a 2 µm tip radius and an interferometer with numerical aperture (NA) 0.55 may agree on *Sa* for one surface yet diverge dramatically for another. The spectral consequences of stylus-tip geometry have been analysed since the early simulations of Li and Devries [4] and the tip-curvature spectral analyses of Wu [5]; the instrument transfer function (ITF) of optical profilometers has been treated analytically by de Groot [6] and measured experimentally with binary pseudo-random calibration artefacts [7]. However, the two measurement principles have not been compared within one consistent $k$-space framework. Instrument bandwidth has been discussed qualitatively for surface measuring instruments [8], and the analytic PSD formalism and systematic spectral measurement errors have been catalogued in the lithography context [9], [10]; neither line of work, however, provides paired transfer functions for the tactile–optical comparison pursued here. The central unresolved question addressed here is: *given a surface with a known PSD, what does each instrument actually measure as a function of spatial frequency?*

## II. PROPOSED CONTRIBUTION

Every surface topography measurement is represented as a linear operator acting on the true height field $h(x, y)$, followed by additive noise:

$$\mathcal{M}[h(x,y)] = \mathcal{F}^{-1}\left[H(f_x, f_y) \cdot \mathcal{F}[h(x,y)]\right] + \eta(x,y), \tag{1}$$

where $\mathcal{F}$ is the two-dimensional Fourier transform, $H(f_x, f_y)$ is the instrument transfer function, and $\eta$ is measurement noise.

For the stylus, the transfer function arises from morphological dilation with a paraboloid tip of radius 2 µm (cone half-angle 22.5°) followed by raster sampling at 200 nm. In the small-slope limit, dilation preserves the height field to first order and adds a curvature bias $(r_{\mathrm{tip}}/2)\langle|\nabla h|^2\rangle$: the empirical power transfer is therefore approximately flat across the passband rather than a steep Gaussian tip filter. For the optical system, the transfer function is the diffraction-limited incoherent optical transfer function (OTF) of a circular aperture,

$$H_{\mathrm{optical}}(f) = \frac{2}{\pi}\left[\arccos(\tilde{f}) - \tilde{f}\sqrt{1-\tilde{f}^2}\right], \tag{2}$$

where $\tilde{f} = \lambda f/(2\,\mathrm{NA}) < 1$, with wavelength 550 nm and NA = 0.55, giving a diffraction cutoff $f_c = 2\,\mathrm{NA}/\lambda = 2000\,\mathrm{mm}^{-1}$. Phase wrapping, one-dimensional row unwrapping with a 1 % error probability, and additive camera noise complete the optical chain.

From the pair of transfer functions, the analysis computes the spectral information loss $I(f)$ =

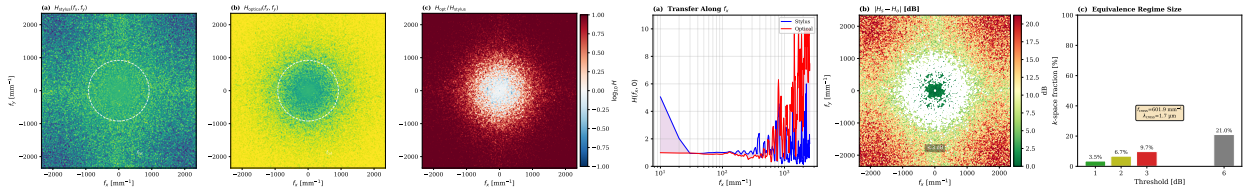


Fig. 1. Left: simulated instrument transfer functions in $k$-space — stylus (flat with curvature-bias excess), optical (diffraction-limited), and their ratio. Right: crossover analysis — the 3 dB equivalence boundary and the equivalence fraction as a function of threshold.

$\log_{10}[\mathrm{PSD}_{\mathrm{true}}/\mathrm{PSD}_{\mathrm{measured}}]$, the measurement anisotropy tensor, and the $k$-space equivalence regime $\Omega_{\mathrm{eq}}(\Delta)$, defined as the frequency set where the two transfer functions agree to within $\Delta$ (in decibels). Information-theoretic diagnostics — mutual information under a Gaussian surface prior, the reconstructability index $\rho = 1 - \mathrm{MMSE}/\sigma^2$, and the capacity–uniformity product — are computed with all estimator conventions documented. A metrological extension of the classical Fourier uncertainty is proposed and evaluated as a falsifiable benchmark,

$$\Delta x_{\mathrm{eff}} \cdot \Delta f_{\mathrm{eff}} \geq \frac{1}{4\pi}\left(1 + \frac{1}{\overline{\mathrm{SNR}}}\right), \tag{3}$$

where $\Delta x_{\mathrm{eff}}$ and $\Delta f_{\mathrm{eff}}$ are the second-moment widths of the point-spread function and of the noise-truncated transfer function, and $\overline{\mathrm{SNR}}$ is the area-averaged signal-to-noise ratio; the expression is anchored to the classical limit $1/(4\pi)$ as $\overline{\mathrm{SNR}} \to \infty$ and is explicitly presented as an adopted benchmark rather than a proved theorem. Every headline result is checked by an automated verification suite of eighteen tests, including an independent Monte Carlo check of the minimum mean-square error expression.

## III. PRELIMINARY RESULTS

Simulations use self-affine surfaces ($H = 0.7$, $\sigma = 50$ nm) on a $512 \times 512$ grid spanning $100\,\mu\mathrm{m} \times 100\,\mu\mathrm{m}$, with fixed random seeds for reproducibility.

**Transfer functions.** The empirical stylus transfer takes values 2.2, 0.91, 0.74 and 0.57 at 17, 292, 567 and 843 $\mathrm{mm}^{-1}$, crossing 3 dB at $\approx 1049\,\mathrm{mm}^{-1}$; the low-frequency excess is the curvature bias, and the Gaussian-tip idealisation $\exp(-2\pi^2 r^2 f^2)$ common in the literature predicts $\sim 10^{-3}$ at $292\,\mathrm{mm}^{-1}$, two orders of magnitude below the simulated value. The optical transfer follows the incoherent OTF to its cutoff at $2000\,\mathrm{mm}^{-1}$ (Fig. 1).

**Equivalence regime.** The modalities agree to within 3 dB over $9.6\,\% \pm 0.3\,\%$ of the measured $k$-space ($n = 30$ ensemble, $N = 512$ grid), with a crossover frequency $f_{\mathrm{cross}} = 454/\mathrm{mm} \pm 166/\mathrm{mm}$ (feature wavelength $\approx 2.2\,\mu\mathrm{m}$). Below the crossover the stylus dominates; above it the optical system retains information up to its diffraction cutoff. A 180-point parameter sweep shows the crossover depends only weakly on the tip radius ($f_{\mathrm{cross}} \propto r_{\mathrm{tip}}^{-0.06\pm0.01}$).

**Accuracy and scaling.** The stylus achieves an rms error of 16.3 nm against the optical 29.3 nm, despite the optical system's higher nominal cutoff: unwrapping errors, not bandwidth, limit the optical accuracy. The sweep is approximately described by an empirical power law $\mathrm{RMSE}_s/\mathrm{RMSE}_o \propto (f_{c,\mathrm{opt}}/f_{c,\mathrm{stylus}})^{1.90\pm0.13}$ (in-sample $R^2 = 0.53$; blocked cross-validation $R^2 = 0.52$), which is reported as an order-of-magnitude descriptor rather than a precise predictor.

### A. Information-Theoretic Diagnostics

Treating the profilometer as a Gaussian channel [11], [12]—the surface a stationary Gaussian field and the measurement a filtered observation with additive noise—the mutual information between surface and measurement decomposes per spatial frequency,

$$I(h;y) = \frac{1}{2}\sum_{\mathbf{f}} \log_2\left[1 + \frac{|H(\mathbf{f})|^2\, S_h(\mathbf{f})}{S_\eta(\mathbf{f})}\right], \tag{4}$$

and the minimum mean-square reconstruction error takes the closed form $\mathrm{MMSE} = \sigma^2 - \iint |H|^2 S_h^2/(|H|^2 S_h + S_\eta)\, d^2 f$, verified against Monte Carlo Wiener filtering in the verification suite. The reconstructability index $\rho = 1 - \mathrm{MMSE}/\sigma^2$ then favours the stylus at the baseline ($\rho_{\mathrm{stylus}} = 0.79$ versus $\rho_{\mathrm{optical}} = 0.64$), and the per-band information loss is lowest at low frequencies (0.071 decades) and highest near the sampling scale (0.446 decades). The total mutual-information ranking is estimator-dependent (ratios of 0.7–1.6 across defensible noise-floor estimators) and is therefore not claimed as robust; only the per-band picture is reported as an estimator-independent result.

**Surface-class stress test and third modality.** Six surface classes (Gaussian self-affine, anisotropic ground, turned, laser-textured, periodic grating, and biological-like; $n = 5$ realisations per class) were analysed to check whether the metrics generalise beyond the calibration class. The reconstructability index is stable across the random rough classes (CV = 0.16) and saturates at unity for strongly deterministic surfaces, where it becomes surface-limited rather than instrument-limited; the optical system's near-zero $\rho$ on the pure periodic grating is correctly explained by the concentration of spectral power at a single $k$-space point. Extending the framework to scatterometry via the Rayleigh–Rice BSDF-to-PSD relation adds a third, flat-response channel spanning $10\,\mathrm{mm}^{-1}$ to $3760\,\mathrm{mm}^{-1}$ and confirms the complementarity of the three modalities; the multi-instrument fusion of the three estimates is retained as a conceptual outlook (see Section IV). The measurement anisotropy is small for both modalities ($\alpha_s = 0.0095 \pm 0.0037$

*Table I. Key verified quantities of the study (all from the committed simulation pipeline and its verification suite).*

| Quantity | Stylus | Optical |
|---|---|---|
| Transfer at 292 $\mathrm{mm}^{-1}$ (empirical power) | 0.91 | follows incoherent OTF |
| 3 dB point | $\approx 1049\,\mathrm{mm}^{-1}$ | $2000\,\mathrm{mm}^{-1}$ (cutoff) |
| Baseline rms error | 16.3 nm | 29.3 nm |
| Anisotropy $\alpha$ ($n = 30$) | $0.0095 \pm 0.0037$ | $0.0013 \pm 0.0006$ |
| Reconstructability $\rho$ (baseline) | 0.79 | 0.64 |

versus $\alpha_o = 0.0013 \pm 0.0006$, i.e. the stylus exceeds the optical value by a factor of about seven), and the spectral information loss increases mildly with frequency (local log–log slope $\approx 0.4$ over $50\,\mathrm{mm}^{-1}$ to $1500\,\mathrm{mm}^{-1}$).

**Public-data cross-checks.** The optical channel model was benchmarked against the public, CC-BY-4.0 WLI dataset of Hirose et al. [13] (Zenodo 10.5281/zenodo.15750880): fitting only a background PSD amplitude, its slope, and one instrument parameter (the cutoff scale $c$) reproduces the measured instrument transfer functions with $c = 1.15 \pm 0.01$ (Fig. 2). The stylus channel admits no comparable paired dataset; as a geometric cross-check, the dilation simulator reproduces the classical parallel-curve radius shift on the published ball-standard calibration profiles of Aksulu [14] within 3 % to 5 % for tip radii of 2 µm to 50 µm, although the published profiles' scatter (40 µm to 120 µm) exceeds the tip scale, so this check is geometric rather than tip-resolving.

**Gap-closure roadmap.** Because the optical OTF is inherently low-pass while the stylus transfer is nearly flat at low frequencies, optical profilometry cannot fully dominate the stylus in $k$-space, but near-total equivalence is attainable. A 432-point sweep ($N = 256$ grid; its own baseline is 48 %) shows that a 100× noise-floor reduction, NA of 0.7–0.9, shorter illumination wavelengths, and two-dimensional phase unwrapping expand the equivalence fraction from the ensemble baseline of 9.6 % to 87 % within the sweep. A physics-limited residual gap remains at the lowest spatial frequencies, where the stylus remains the primary standard.

## IV. POTENTIAL APPLICATIONS

The framework gives practitioners quantitative criteria for instrument selection. For feature wavelengths between 0.5 µm to 1.5 µm (frequencies $670\,\mathrm{mm}^{-1}$ to $2000\,\mathrm{mm}^{-1}$), the optical measurement is preferred; in the crossover band (1.5 µm to 2 µm) the two measurements carry comparable information and can be averaged; for wavelengths above 2 µm the stylus is preferred. Below 0.5 µm — beyond the optical diffraction cutoff — neither modality resolves surface features in the modelled passband. The crossover frequency serves as a trust boundary for cross-validation of real measurements. The gap-closure roadmap prescribes four instrument improvements with the highest equivalence gain for optical systems: (i) a 100× reduction of the optical noise floor (cooled sCMOS detectors, brighter illumination and multi-frame averaging); (ii) higher numerical apertures (0.7–0.9); (iii) shorter illumination wavelengths (400 nm to 450 nm); and (iv) two-dimensional quality-guided phase unwrapping in place of one-dimensional row unwrapping. For instrument qualification, the transfer-function framework complements scalar benchmarks: two instruments with equal rms error can differ markedly in the spatial-frequency bands they preserve, and the equivalence boundary provides a quantitative cross-validation rule. A $k$-space fusion of stylus, WLI and scatterometry, weighted by local information uncertainty, is presented as a conceptual outlook rather than a quantitative validation, since the simulated scatterometry PSD is not yet on the same absolute scale as the height PSD.

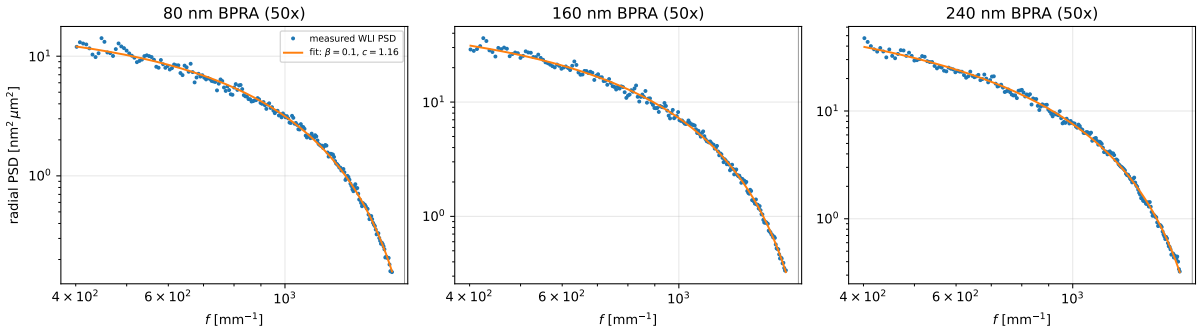



Fig. 2. Public-data validation of the optical channel model: measured WLI radial PSDs (points) of three binary-pseudo-random-array samples (Hirose et al., CC-BY-4.0) and the fitted model (lines) with fitted cutoff scale $c$; the vertical line marks the nominal monochromatic cutoff $2\mathrm{NA}/\lambda$.

## V. CONCLUSIONS

A paired, first-principles $k$-space comparison of stylus and optical profilometry was developed and verified. Its central, reproducible findings are: (i) the stylus transfer is approximately flat with a curvature-bias excess and rolls off only near the sampling scale, refuting the classical Gaussian-tip picture; (ii) the optical transfer follows the incoherent OTF and reproduces measured public WLI transfer functions; (iii) the modalities are interchangeable over $9.6\,\% \pm 0.3\,\%$ of $k$-space with a quantitative crossover at $\approx 454\,\mathrm{mm}^{-1}$; and (iv) the gap between the modalities can be narrowed to 87 % equivalence with four prescribed instrument improvements. All results are computational and are offered with a falsifiable experimental roadmap; an experimental realisation with calibrated artefacts remains the decisive next step. The complete simulation framework, including the eighteen-test verification suite and the audit scripts behind every reported number, is available in a public repository [15] (release v1.0.1, Zenodo DOI 10.5281/zenodo.23493876).

## AUTHORS' CONTRIBUTION

Dawid Kucharski: conceptualisation, methodology, software, formal analysis, investigation, writing — original draft. Michał Jakubowicz: validation, visualisation, writing — review and editing.

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